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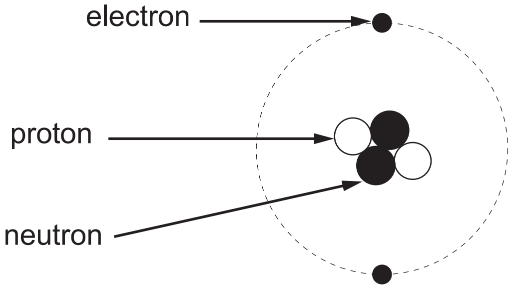
3. A chemistry student correctly draws and labels a helium atom.



(a) An alpha particle is a helium nucleus. State the difference between the structure of an alpha particle and a helium atom. [1]

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(b) A nucleus of polonium-218 ($^{218}_{84}\mathrm{Po}$) decays by an alpha (α) emission.

$$^{218}_{84}\mathrm{Po} \longrightarrow \mathrm{X} + {}^{4}_{2}\alpha$$

A short time later the resulting nucleus X decays by a beta (β) emission.

$$\mathrm{X} \longrightarrow \mathrm{Y} + {}^{0}_{-1}\beta$$

Nucleus Y then decays by a beta (β) emission.

$$\mathrm{Y} \longrightarrow {}^{214}_{84}\mathrm{Z} + {}^{0}_{-1}\beta$$

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The following table gives information about some elements, their symbols and proton numbers.

Element	Symbol	Proton number
radium	Ra	88
francium	Fr	87
radon	Rn	86
astatine	At	85
polonium	Po	84
bismuth	Bi	83
lead	Pb	82

(i) Use the information above to complete this table. [4]

	Element	Nucleon number
X	....	....
Y	....	....

(ii) State which of X, Y or Z is an isotope of polonium (Po). .... [1]

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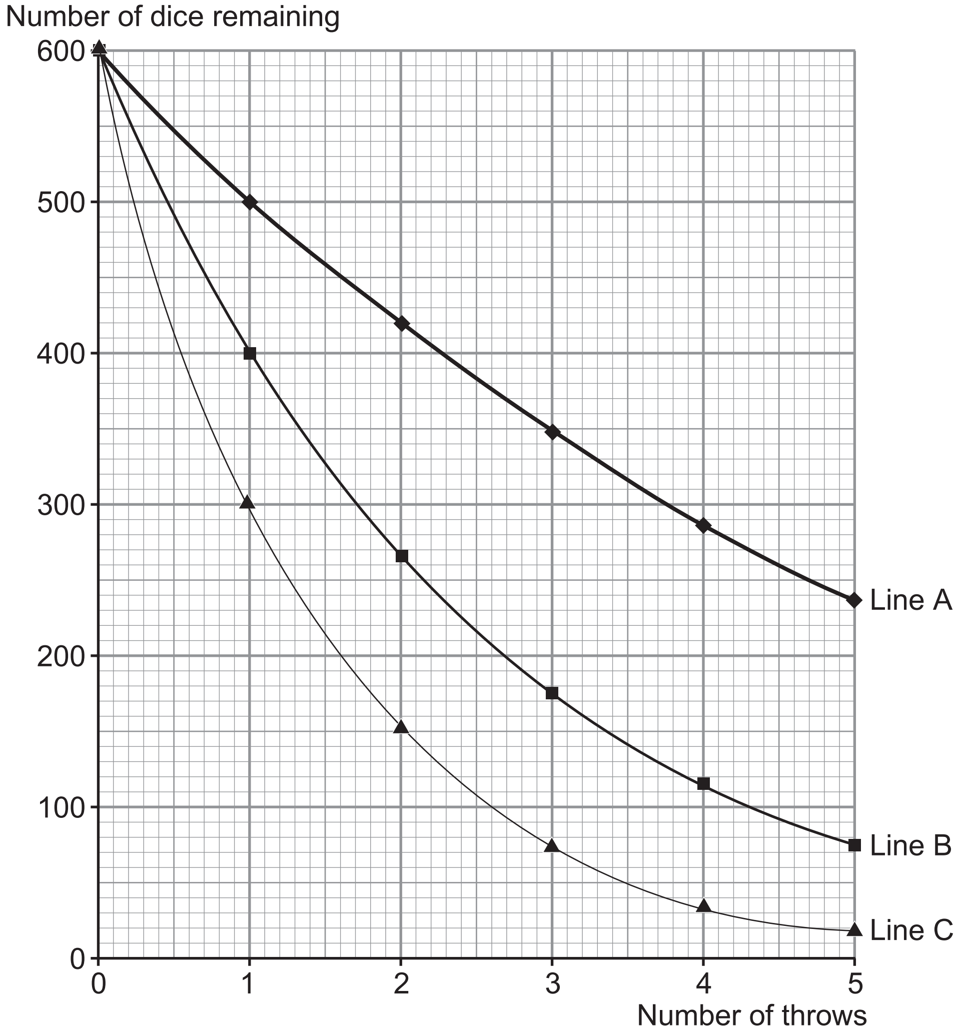
5. A physics teacher uses 600 identical red dice with her class to model the decay of different radioisotopes. To collect her first set of data she throws the 600 dice. She then removes all of the **sixes** that are face up. She then counts the remaining dice and repeats the process for a total of 5 throws.

In her second demonstration she repeats the experiment but removes all of the **fives** and **sixes** that are face up.

In her third demonstration, she removes all of the **fours**, **fives** and **sixes** that are face up.

She records her data on a spreadsheet. This quickly produces a graph for her class to see. The graph is shown below.



(a) (i) Explain why the teacher chooses to use a large number of dice for her demonstration. [2]

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(ii) Using only information from the graph tick (✓) the **three** correct statements. [3]
Space for working:

Line A represents the data from the third demonstration. ☐

Line B shows the equivalent of three half-lives in 5 throws. ☐

Line C could also represent the same demonstration modelled with 600 coins. ☐

Line A represents the radioisotope that decays at the quickest rate. ☐

Line B has an activity that is $\frac{1}{8}$ of its original after 5 throws. ☐

Line C represents the longest half-life. ☐

(iii) The teacher suggests to her class that an improved model of radioactive decay would be to include 20 blue cubes, along with the 600 red dice, at the start. The blue cubes **would not** be removed during the demonstration but would be counted each time.

I. State what physical quantity the blue cubes represent. [1]

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II. State how the presence of the blue cubes affects **line A** on the graph. [1]

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(b) Strontium-93 (Sr-93) is a radioactive source which decays with a half-life of 8 minutes.

(i) A sample of Sr-93 has a mass of 68.0 mg. Calculate the mass of undecayed Sr-93 left after 56 minutes. Show your working and give your answer to **1 significant figure**. [3]

Mass = ... mg

(ii) A scientist works with Sr-93 which decays by beta and gamma emission. When he is not using the Sr-93, explain why, for his own safety, he has to store it in a lead box. [3]

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2. Radiotherapy is used to treat cancer. Three types of radiotherapy are described below.

Brachytherapy is a type of internal radiotherapy. It involves putting a sealed radiation source inside the cancerous growth. The radioisotope used emits low energy gamma rays. An isotope of iodine (iodine-125) can be used to treat prostate cancer.

Unsealed source radiotherapy also uses radioactive substances to treat cancer. These are introduced into the body by injection or ingestion. Iodine-131 is injected into a patient to treat thyroid cancer.

External radiotherapy is different from the methods described above. It is given as a series of short, daily treatments in the radiotherapy department using high energy gamma rays.

Information about some isotopes of iodine is given below.

Iodine-123 has a half-life of 13 hours and emits gamma.
Iodine-125 has a half-life of 59 days and emits gamma.
Iodine-128 has a half-life of 25 minutes and emits beta.
Iodine-129 has a half-life of 15 000 000 years and emits beta and gamma.
Iodine-131 has a half-life of 8 days and emits beta and gamma.

(a) Explain why iodine-123 is unsuitable to treat prostate cancer. [2]

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(b) Explain why iodine-131 is more suitable to treat thyroid cancer than iodine-128. [2]

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(c) Patients are told that, after treatment with iodine-131, small amounts of radiation from their body may trigger radiation monitors until the activity has dropped to one thousandth $\left(\frac{1}{1000}\right)$ of its initial value. The patients are told this will occur 80 days after treatment. Explain with the aid of a calculation whether 80 days is long enough. [3]

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Answer **all** questions.

1. (a) Background radiation occurs due to the decay of an unstable nucleus releasing alpha, beta or gamma radiation.

Complete the table below to state what each type of radiation is. [3]

Type of radiation	Symbol	What it is
Alpha	${}^{4}_{2}\alpha$	....
Beta	${}^{0}_{-1}\beta$	....
Gamma	γ	....

(b) A teacher demonstrates how to determine the background radiation count in her laboratory.
She uses a radiation detector and measures 18 counts in 30 seconds.

(i) Determine the background radiation count in counts per second. [1]

Background radiation = ... counts per second

(ii) Suggest **two** ways that the teacher could improve her measurement of background radiation. [2]

1. ....

2. ....

(iii) Explain why the background radiation count is much higher in some parts of the country than others. [2]

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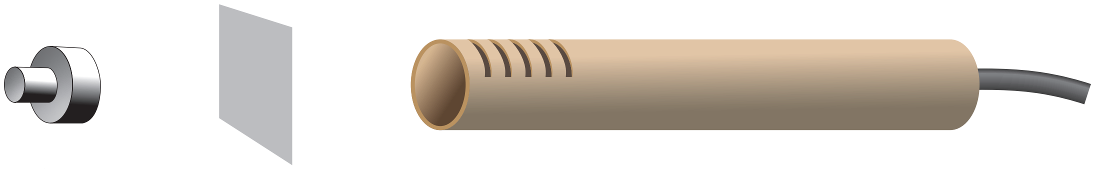
(c) The teacher now demonstrates an experiment to identify the radiation emitted by different sources.
The count rate is measured, first with no absorber present and then with paper and aluminium absorbers separately.



The results are shown in the table below. They are corrected for background radiation.

Source	Count rate (counts per second)		
	No absorber	Paper	Thin aluminium
1	312	313	312
2	389	57	0

(i) Write **yes (Y) or no (N)** in each box to show which type(s) of radiation is (are) emitted by each source. [4]

	Alpha	Beta	Gamma
Source 1			
Source 2			

(ii) Explain **one** change the teacher could make to extend the investigation to confirm to the class the conclusions made. [2]

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4. A class investigated radioactive decay using 50 dice, each with 8 sides.

They rolled the dice and removed any which landed with an 8 facing upwards, to represent a decayed nucleus.

They recorded the number of dice remaining.

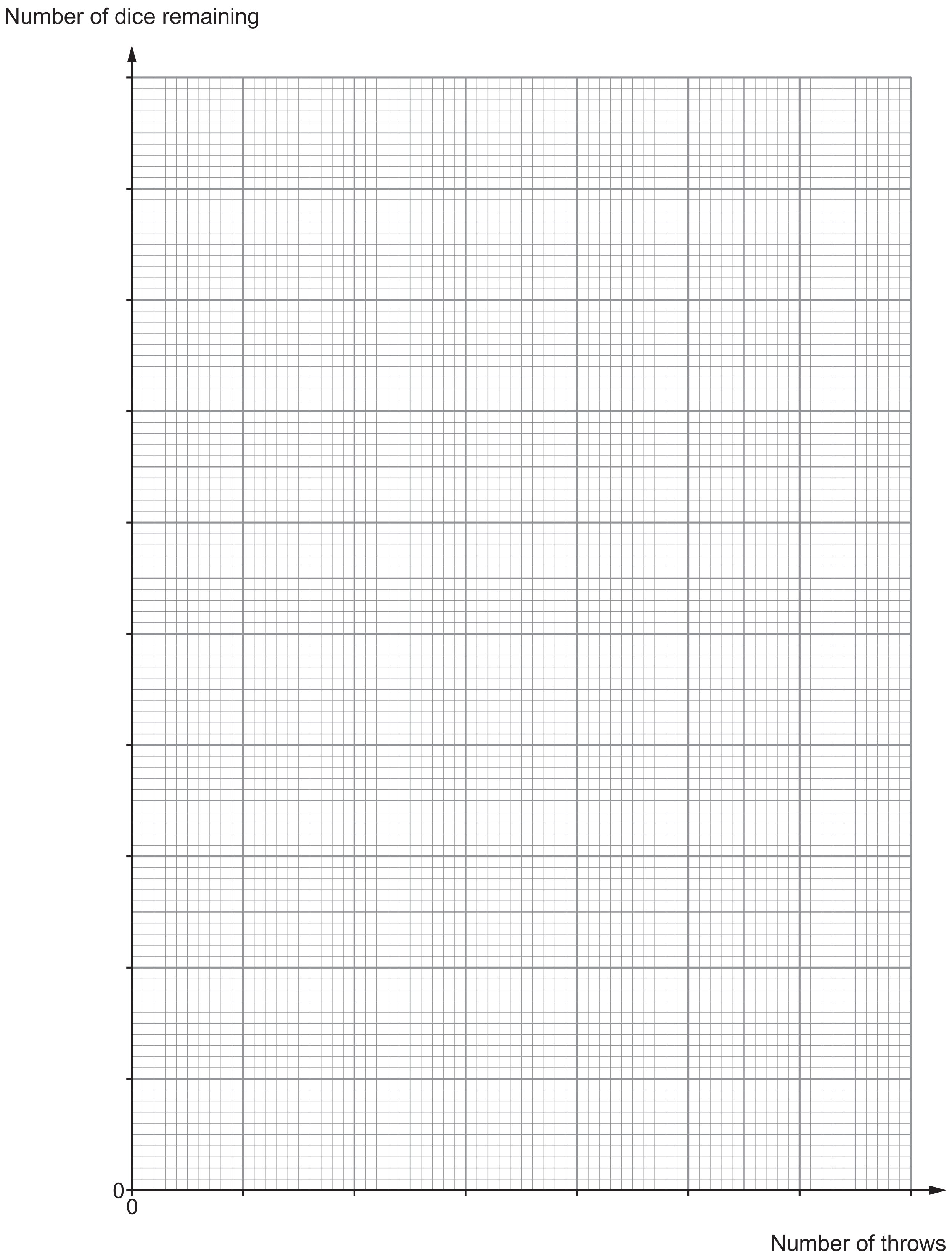
They rolled the remaining dice and again removed any which landed with an 8 facing upwards.

This was repeated until they had rolled the dice 10 times in total.

Some of the results are shown in the table below.

Number of throws	Number of dice remaining
0	50
2	37
4	29
6	23
8	18
10	15

(a) (i) Plot the data on the grid opposite and draw a suitable line. [3]


Number of dice remaining
0
0
Number of throws

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(ii) Use your graph to determine how many dice were removed on the **3rd throw**. [2]

Number of dice removed = ...

(iii) Determine how many throws it took for the initial number of dice to halve. Show how you obtained your answer on the graph. [2]

Number of throws = ...

(iv) If the experiment was repeated with 1 000 dice, use your results to predict how many throws it would take to reduce them to 125. [2]

Number of throws = ...

(b) Explain why it is preferable to use a larger sample size in this experiment. [2]

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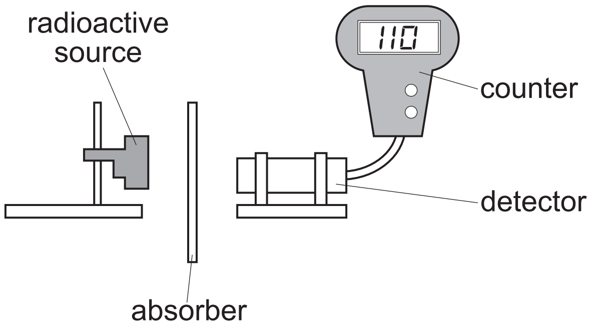
Answer **all** questions.

1. A teacher uses the apparatus below to demonstrate the penetrating properties of alpha, beta and gamma radiation.



(a) The teacher explains that there is a possibility of exposure to radiation from the source. Complete the risk assessment below. [2]

Hazard	Risk	Control measure
Nuclear radiation is ionising	....	....

(b) After the experiment the teacher gives the students some data about the radioactive source, cobalt-60, to analyse.
The data are given in the table below.

Absorber	Count rate (counts per second)
no absorber	256
paper	256
aluminium	110
lead	50

Use the data to answer the following questions.

(i) Explain how the data show that cobalt-60 does not emit alpha particles. [1]

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(ii) Explain how the data show that cobalt-60 emits beta and gamma radiation. [2]

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(iii) The teacher tells the class that counts due to background radiation are included in the results in the table.

I. State **one** cause of background radiation. [1]

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II. State how the results in the table should be corrected for background radiation. [1]

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6. In one example of the process of nuclear fission of uranium-235, it is suggested that isotopes of caesium and rubidium and a number of neutrons ($^{1}_{0}n$) are produced.

The table gives information about the particles in the nuclei of the isotopes named above.

Element	Symbol	Number of protons in the nucleus	Number of neutrons in the nucleus
uranium	U	92	143
caesium	Cs	55	85
rubidium	Rb	37	55

(a) Write a balanced nuclear equation for the fission decay of uranium into caesium and rubidium. [4]

$$^{\dots\dots}_{\dots\dots}U \;+\; ^{1}_{0}n \longrightarrow \dots\dots\dots\dots \;+\; \dots\dots\dots\dots \;+\; \dots\dots\dots\; ^{1}_{0}n$$

(b) (i) Explain the purpose of the moderator in a fission reactor. [2]

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(ii) Explain how the rate of reaction can be increased in a nuclear reactor. [3]

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(c) The caesium isotope produced in this fission reaction has a half-life of 64 s.
The rubidium isotope has a half-life of 4 s.
At time, $t = 0$ the mass of caesium present is 131 072 g.
The same mass of rubidium is also present.

(i) Calculate the mass of caesium present after 64 s. [1]

mass of caesium = ... g

(ii) I. Calculate how many half-lives of rubidium occur in 64 s. [1]

number of half-lives = ...

II. After 32 seconds, 512 g of rubidium remains.
Calculate the mass of rubidium remaining after 64 s. [3]

mass = ... g

(iii) Initially the ratio of mass of caesium to mass of rubidium was 1:1.
Explain how this ratio changes with time. [2]

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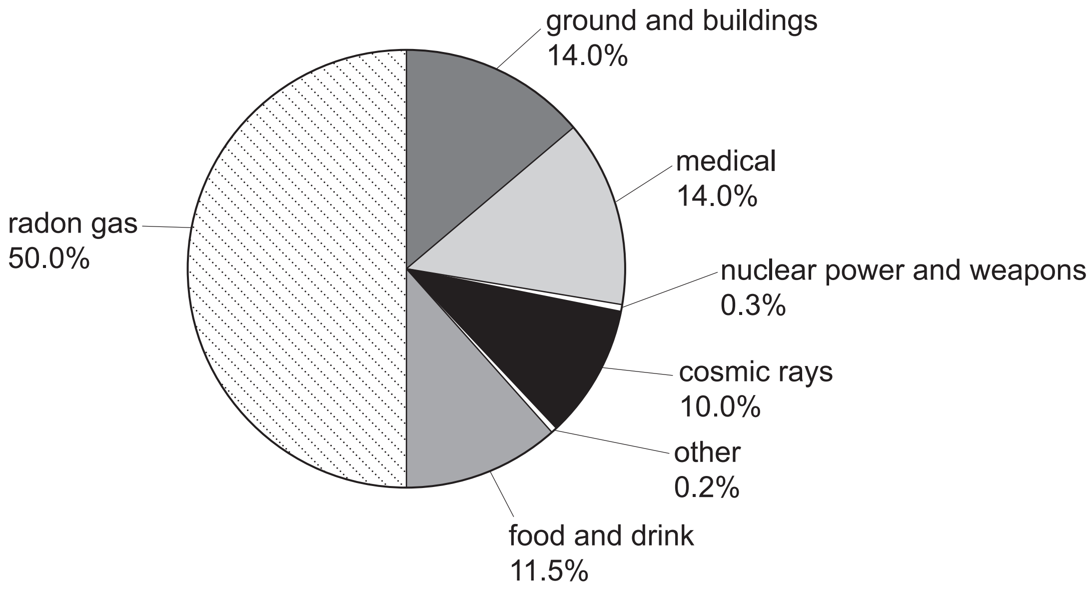
Answer **all** questions.

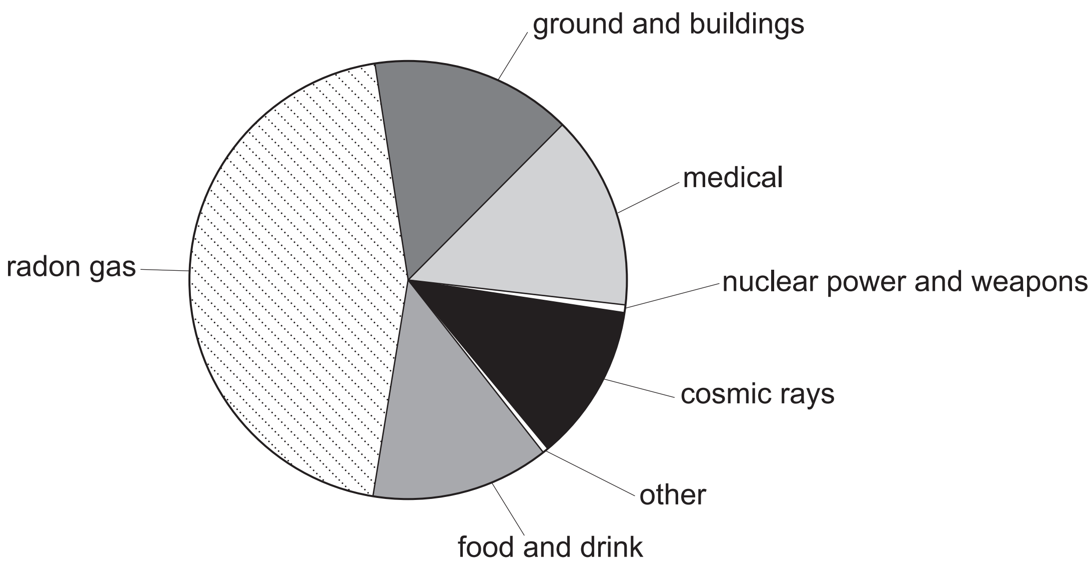
1. A group of students is investigating background radiation.

(a) They find the pie charts below, which show the background radiation in 3 different locations, A, B and C, in the UK.

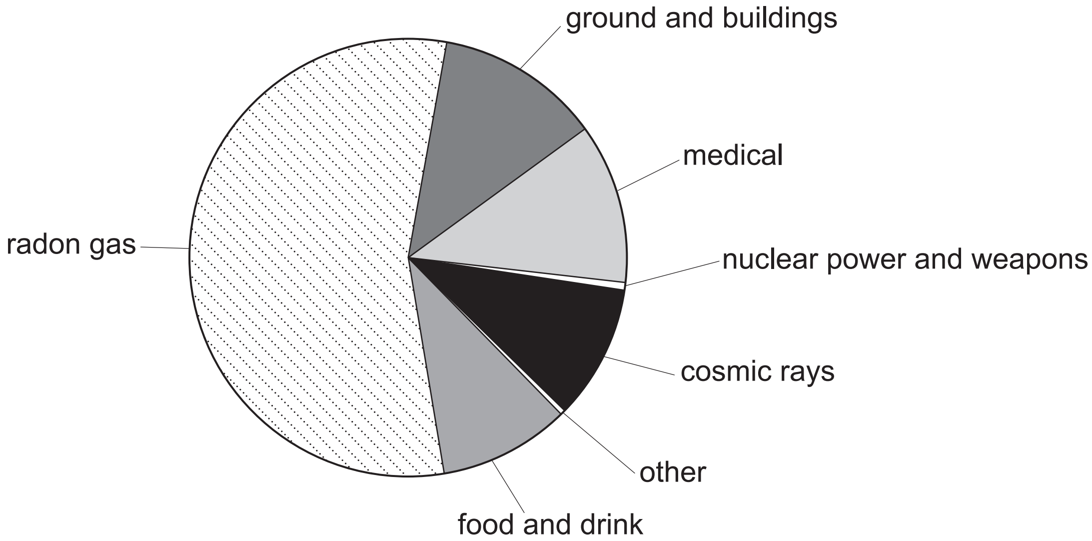
Location A



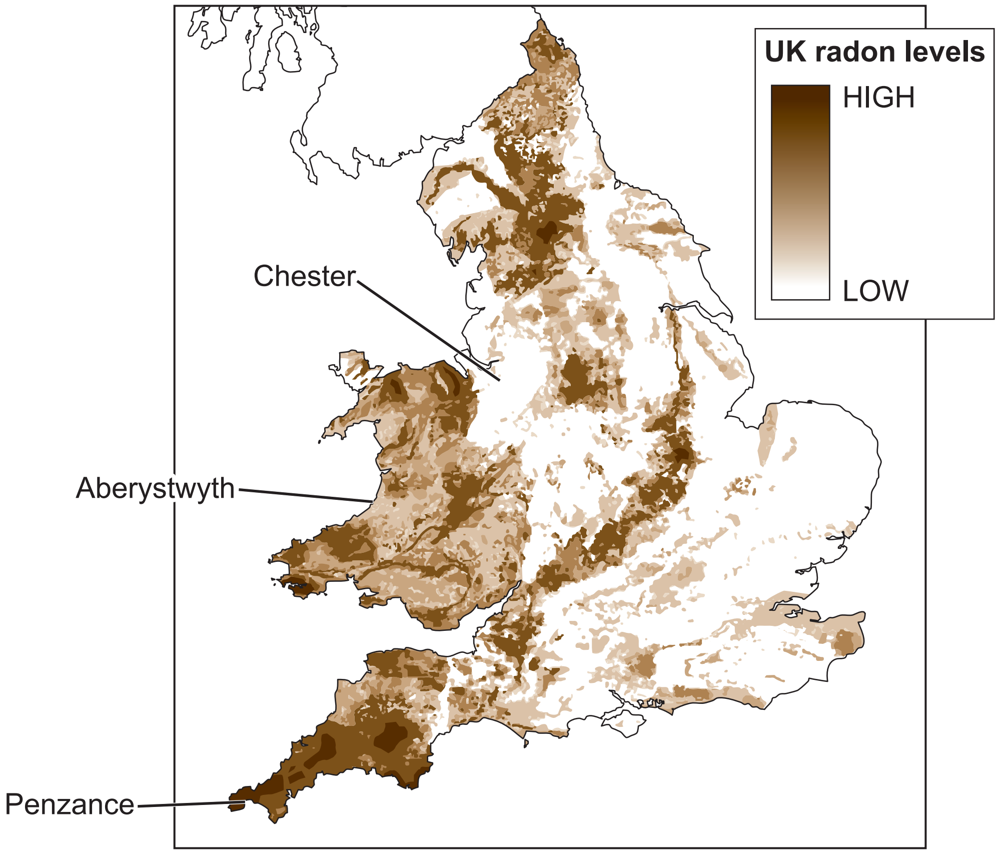
Location B



Location C



The map below shows radon levels across the UK.
The 3 locations, A, B and C are shown on the map.
The darker the area on the map the higher the levels of radon.



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(i) Adam studies the diagrams and concludes that **location A** must be Aberystwyth. Explain whether you agree. [2]

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(ii) Their teacher measures the background count in **location A**. The teacher records 30 counts in 60 seconds.

I. Determine the count rate in counts per second. [1]

count rate = ... cps

II. Use information from the pie chart to determine how many of the 30 counts are due to radon gas. [1]

counts due to radon = ...

(iii) Chloe states that people are more at risk from man-made sources of background radiation than natural sources.
Use data from the pie chart for **location A** to explain whether you agree. [2]

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(b) The table shows how much radiation people receive from some different sources.

Source of radiation	Radiation received (units)
mean background radiation (per year)	2.7
8 hour flight from London to New York	0.09
dental X-ray	0.005
chest X-ray	0.014
CT scan of head	1.4
CT scan of chest	6.6
worker in a nuclear power station (per year)	0.18

Workers in nuclear power stations have their exposure to radiation carefully monitored.

If they receive a total of **20 units** of radiation from all sources including background radiation in one year, they can no longer work with radiation.

(i) Sophia works in a nuclear power station.
In one year, she flies from London to New York and back.
She also has a dental X-ray and a CT scan to her chest.
Sophia is worried about the level of radiation she has been exposed to.

Use data to explain whether it is still safe for her to work with radiation. [2]

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(ii) Jack states that workers in a nuclear power station are exposed to more radiation in one year than airline pilots flying on the London to New York route.

Use data to explain whether he is correct. [2]

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5. Radioactive isotopes have a wide variety of uses.

(a) (i) State what is meant by the term ‘isotopes’. [2]

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(ii) State why some isotopes are radioactive. [1]

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(b) One use of radiation is in a smoke detector.

The radiation ionises the air inside the detector and causes a current that is detected.
If smoke particles are present they block the radiation so that a current is no longer detected.
This sets off the alarm.

The table gives information about different radioisotopes.

isotope	half-life	type of emitter
americium-241	432 years	alpha
radium-224	3.6 days	alpha
caesium-137	30 years	beta and gamma

Explain why americium-241 is suitable for this use but radium-224 and caesium-137 are not. [3]

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(c) Brachytherapy is a form of radiotherapy used in treating prostate cancer. Pellets containing radioactive material are planted inside the prostate. One patient has 75 pellets containing iodine-125 planted in his prostate. Each pellet has an initial activity of 16 MBq.

(i) The iodine-125 has a half-life of 60 days. Calculate the time taken for the total activity of the iodine-125 in the 75 pellets to drop to 37.5 MBq. [3]

time = ... days

(ii) Iodine-125 is a gamma emitter. For 60 days after his treatment the patient is warned not to stay too close to people for long periods of time. Explain why the gamma radiation may pose a risk to other people. [2]

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